October 5–9, 2026

Feedback in Practice

Turning goals into student action through evidence, feedback, and the next lesson

THIS WEEK'S PLC

Feedback → Student Action → New Evidence

Our PLC sequence protects time for teachers to plan feedback students can use, rehearse the learning loop, and decide what evidence will guide the next lesson.

LESSON INTERNALIZATION & GOAL SETTING

From Goals to a Practical Educator Plan

This week, teams connected Student Learning and Professional Practice goals to one upcoming learning experience. The emphasis was practical and concrete: identify an instructional move, plan how students will experience it, and name the evidence that will show whether the move supported learning.

Teams also considered the difference between a comment that evaluates work and feedback that gives students a productive next step. The goal is not simply to identify an error; it is to make the next revision, explanation, or application visible and doable.

> **Shared focus:** useful feedback does not do the thinking for students. It gives them a specific next action—revise, correct, extend, explain, compare, or apply—to move their learning forward.

01 · EDUCATOR PLAN

Make Goals Visible in a Lesson

Teams selected one upcoming lesson and connected their goals to practical actions, supports, materials, timing, and a first implementation step.

- Choose the task or student-work moment.
- Plan one focused instructional move.
- Name the feedback-for-improvement loop.

02 · FEEDBACK ROUTINE

Plan Feedback That Prompts Action

Teachers identified where student thinking becomes visible and planned a feedback message students can use immediately.

- Locate the evidence moment.
- Name what students will do next.
- Protect cognitive demand and access.

03 · EVIDENCE CHECK

Rehearse, Notice & Adjust

Course-alike teams rehearsed prompts, looked for student action, and set the next evidence check.

- Use available evidence when possible.
- Calibrate the feedback-to-action routine.
- Record one next-lesson adjustment.

Learning goal
What should students know or be able to do?

Evidence moment
Where will student thinking become visible?

Specific feedback
What can students use immediately?

Student action
How will they revise, explain, or apply?

New evidence
What will the team notice next?

FEEDBACK IN ACTION · MR. JONATHAN WAY

Feedback That Invites Students Back Into Their Work

We appreciate Mr. Way's thoughtful use of feedback practices that encourage students to interact with feedback in Google Classroom and continue improving their work.

The focus is a learning routine in which students read feedback, make a revision, and use the revised work to strengthen their scientific reasoning.

> A respectful feedback process keeps the emphasis on student growth, clarity, and revision while protecting individual work.

ACTION-ORIENTED FEEDBACK

What Makes Feedback Useful?

CONNECT TO THE GOAL
Refer students back to the learning target, success criteria, or scientific claim they are developing.

NAME ONE NEXT MOVE
Give a manageable action: add evidence, clarify a term, organize an explanation, or revise a claim.

LEAVE THINKING WITH STUDENTS
Use prompts that require students to decide how to improve rather than simply copy an answer.

THE STUDENT RESPONSE LOOP

Read → Revise → Explain

Feedback has the most value when students have time and structure to do something with it.

1. **Read:** identify what is clear and what needs attention.
2. **Revise:** make a visible improvement using evidence or course vocabulary.
3. **Explain:** describe how the revision strengthens the work.

> This loop can support learning in every science course—from laboratory analysis and modeling to written explanations and project work.

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Classroom Spotlight

AP Biology students investigate osmosis, diffusion, and membrane transport through a hands-on model



AP Biology students plan, measure, record, and prepare to analyze evidence from a dialysis-tube model.

AP BIOLOGY · MS. AVA GAGNON

Modeling Movement Across a Membrane

Students planned and carried out an osmosis and diffusion investigation using dialysis tubing as a selectively permeable membrane model. Their work connected a laboratory experience to concentration gradients, water movement, and the exchange of materials in living systems.

By determining what to measure before beginning, groups turned a familiar lab setup into an evidence-based question: what change in mass or concentration would support a claim about molecular movement?

Big question: How do concentration gradients and selectively permeable membranes affect the movement of water and molecules?

FROM PLAN TO EVIDENCE

Design · Measure · Analyze

Before beginning the investigation, student groups developed a procedure, organized laboratory notebooks, and built data tables to record measurements. Groups used labeled solutions, dialysis tubing, balances, pipettes, and graduated cylinders as they prepared model cells and compared conditions.

Students measured initial and final mass, organized inside/outside solution data, and prepared to use those changes as evidence of water movement. The hands-on setup made abstract ideas such as osmosis, diffusion, and hypertonic or hypotonic conditions observable and discussable.

The data table gives students a way to slow down and separate a measurement from an explanation. A mass change is evidence; the scientific work is explaining how the direction of water movement, membrane selectivity, and concentration conditions produced that change.

As groups move from data to explanation, they are also asked to consider how a dialysis-tube model resembles—and differs from—a living cell membrane. That comparison helps students connect the investigation to surface-area-to-volume ratio, exchange across membranes, and cellular homeostasis.

Students used a model to ask: "What evidence would show the direction and rate of water movement?"

FROM MODEL TO ORGANISM

The same reasoning helps students explain why cells must regulate water balance. They can extend the model to predict what may happen when cells encounter different surrounding solutions and evaluate where the model is helpful—and where it is limited.

SCIENCE PRACTICES IN ACTION

Plan · Investigate · Record · Explain

PLAN	Develop a procedure and identify what evidence will be collected before beginning the model investigation.
INVESTIGATE	Use a dialysis-tube model to explore concentration gradients and selective permeability.
RECORD	Organize initial mass, final mass, solution conditions, and measurements in a data table.
EXPLAIN	Use evidence to predict water movement in model cells and living cells.
REFINE	Distinguish what the model shows clearly from what is more complex in an actual cell membrane.

Student thinking to listen for

"The mass changed because water moved across the selectively permeable membrane in response to a concentration difference."

A complete explanation connects the measurement to particle movement, not merely to the direction of mass change.

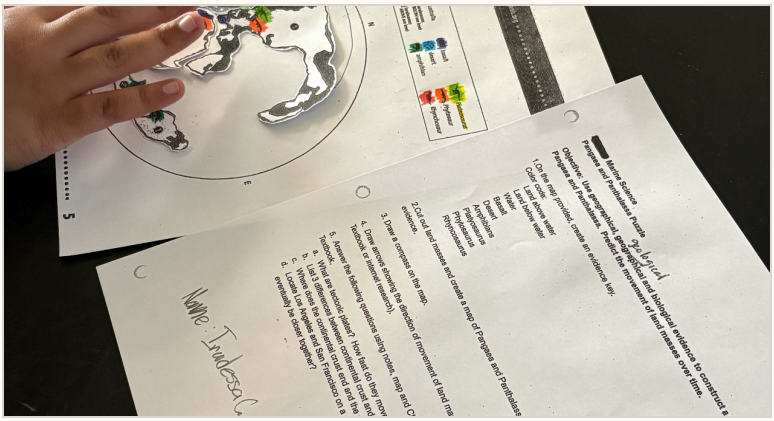
By connecting a model investigation to careful data collection and analysis, students make molecular movement visible through evidence.

Next evidence question: If two groups report different changes in mass, what variables should they revisit before deciding what happened to the model cells—solution concentration, setup, measurement precision, or membrane conditions?

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Classroom Spotlight

Marine Science students use evidence to reconstruct Pangaea and understand plate movement



Students use a color-coded evidence key and land-mass pieces to construct a model of Pangaea and Panthalassa.

HONORS MARINE SCIENCE · MS. PATRICIA GIFFORD

Reconstructing a Changing Earth

Students used the Pangaea and Panthalassa Puzzle to construct a world map showing how land masses may have moved over time. The task asked them to combine geographic, geologic, and biological evidence in a hands-on model of a dynamic Earth.

Rather than treating continental drift as a fact to memorize, students used a map to test possible placements, compare patterns, and explain why multiple lines of evidence create a stronger scientific argument.

> **Big question:** What patterns in fossils, rock, landforms, and seafloor features support the idea that continents were once connected?

EVIDENCE BECOMES A MAP

Piece Together the Story of Plate Motion

Working collaboratively, students positioned land-mass and tectonic-plate pieces while creating an evidence key for fossils, basalt, desert patterns, land above and below sea level, and water. The model invited students to use multiple lines of evidence rather than rely on a single clue.

Students added a compass and arrows to show direction of movement, then used maps, notes, and course resources to revise their models. The puzzle created a shared space for explaining how plate movement, seafloor spreading, and subduction shape Earth's surface over time.

One pattern may suggest an idea, but several connected patterns make the explanation stronger. Matching fossil distributions, rock evidence, and the shape of continental margins provide students with different ways to test a claim about ancient connections.

As students revise their maps, they also practice an important scientific habit: a model is useful when it helps explain evidence, but it can be improved when new evidence or a better explanation becomes available.

> ***Students moved from "Where does this piece fit?" to "What evidence supports that placement and direction of motion?"***

FROM MAP TO MECHANISM

The model opens a pathway from historical evidence to Earth processes. Students can connect the puzzle pieces to a larger explanation of tectonic plates: new crust forms at spreading centers, older crust is recycled at subduction zones, and continents move as part of those plates.

SCIENCE PRACTICES IN ACTION

Observe · Compare · Model · Explain

OBSERVE	Identify fossil, rock, desert, landform, and ocean-floor patterns represented in the evidence key.
COMPARE	Look for patterns across continents that suggest a historical connection.
MODEL	Construct a Pangaea/Panthalassa map and show movement with directional arrows.
EXPLAIN	Use the model and evidence to reason about tectonic plates and changing land masses.
REVISE	Adjust a model when a new pattern, comparison, or explanation calls for a more coherent fit.

Student thinking to listen for

"One clue is interesting, but the connected patterns make a stronger case that these continents were once joined."

That shift from a single observation to a converging-evidence argument is central to scientific reasoning.

Hands-on models give students a meaningful way to test ideas, compare evidence, and explain the forces that continue to reshape our planet.

Next evidence question: How could the age of seafloor rock, the pattern of earthquakes, or the location of plate boundaries provide an additional test of a Pangaea/Panthalassa model?

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Science Corner

The 2026 Nobel Prizes: turning light into a brain tool and chemistry into a mirror-image mystery

NOBEL PRIZE IN PHYSIOLOGY OR MEDICINE · 2026

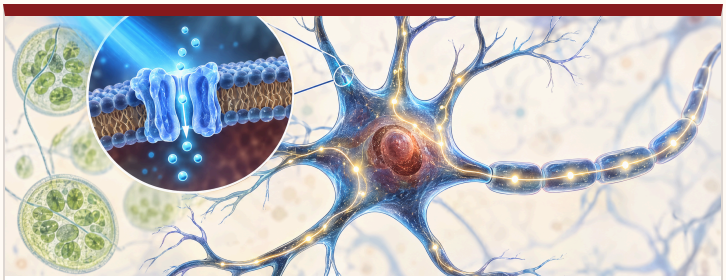
Karl Deisseroth, Peter Hegemann, and Georg Nagel were honored for discoveries concerning light-gated ion channels and optogenetics.

A protein first studied in algae became a precise way to test how selected neurons contribute to brain function.

NOBEL PRIZE IN CHEMISTRY · 2026

Henri B. Kagan and Kenso Soai were honored for discoveries of non-linear effects and autocatalysis in asymmetric organic synthesis.

Their work explains how small differences between molecular mirror images can become amplified in a chemical reaction.



Editorial illustration: optogenetics uses light to control selected nerve-cell activity.

MEDICINE: A LIGHT SWITCH FOR NEURONS

Optogenetics

Peter Hegemann and Georg Nagel discovered channelrhodopsin, a light-sensitive protein from a single-celled alga. Their work established that channelrhodopsin-2 is a directly light-gated, cation-selective channel: after it absorbs light, ions can pass through the cell membrane [1,2].

Karl Deisseroth and colleagues then used the channel as a genetically targeted light switch in neurons. Short pulses of light could reliably control neural spiking on the millisecond time scale—the pace at which neurons communicate [3].

This is more precise than broadly stimulating a whole brain region. Researchers can ask what happens when a particular cell type or circuit is active at a specific time, then compare that result with a carefully designed control. The approach has transformed studies of memory, sensation, behavior, and brain disorders [1,3].

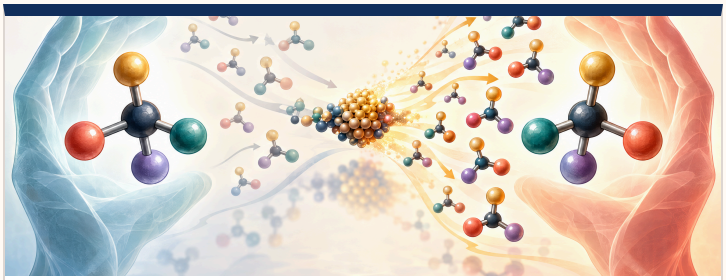
Optogenetics is primarily a research tool, not a general treatment. Yet it illustrates how basic discoveries about algae can become powerful tools for studying the nervous system and for exploring ways to restore light responses in some visual disorders [1].

> ***Key idea: genes identify the target cells; light controls the channel; the resulting change in ion flow changes neural activity.***

HOW A CAUSAL TEST IS BUILT

- Target a chosen population of cells.
- Apply a precisely timed light pulse.
- Measure the resulting neural or behavioral change.

Ask: *Why is a control group essential before claiming that a particular neural circuit caused an effect?*



Editorial illustration: chirality describes molecules that are mirror images, like left and right hands.

CHEMISTRY: CHOOSING ONE MIRROR IMAGE

Autocatalysis & Chirality

Many molecules exist as two mirror-image forms, called enantiomers. They may share the same atoms but fit biological molecules differently—like left and right gloves. Because proteins are built from one molecular "handedness," chemists care deeply about controlling which form a reaction produces [4].

Kagan's work showed that the relationship between the handedness of a catalyst and the handedness of the product can be non-linear: a small difference can lead to a much larger chemical outcome. Soai's reaction revealed a striking form of asymmetric autocatalysis, in which the product helps make more of itself while amplifying an initial imbalance [4,5].

These discoveries gave chemists a model for understanding how one molecular orientation can become dominant. They also help guide the design of selective syntheses for medicines, where choosing the intended enantiomer can be essential for how a compound interacts with a biological target [4,6].

The work does not solve every question about the origin of life's homochirality, but it provides a rare, testable chemical system for studying how tiny molecular differences may be amplified into a large asymmetry [5,6].

> ***Key idea: autocatalysis is a positive-feedback process: the product helps accelerate production of more product, which can magnify a small starting bias.***

WHY CHEMICAL "HANDEDNESS" MATTERS

- Enantiomers can interact differently with a biological target.
- Selective synthesis aims to make the intended orientation.
- Non-linear effects reveal that small catalyst differences can have an outsized result.

Ask: *How can a reaction with the same atoms produce a molecule that behaves differently in a living system?*

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[3] Boyden, E. S., Zhang, F., Bamberg, E., Nagel, G., & Deisseroth, K. (2005). Millisecond-timescale, genetically targeted optical control of neural activity. *Nature Neuroscience, 8*, 1263–1268.

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[5] Soai, K., Shibata, T., Morioka, H., & Choji, K. (1995). Asymmetric autocatalysis and amplification of enantiomeric excess of a chiral molecule. *Nature, 378*, 767–768.

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Why these discoveries connect: Both prize areas begin with a basic scientific question—how does a protein respond to light, and how can one molecular mirror image become dominant? In each case, careful experiments transformed a puzzling observation into a tool for asking new questions about living systems.

> **Science conversation starter:** Which discovery in this article began with a basic observation, and what additional experiment would you design to test its mechanism or application?

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Science Team Recognition

One department, many contributions, and a shared commitment to excellent science learning

THANK YOU
SCIENCE TEAM

A DEPARTMENT THAT WORKS TOGETHER

Preparation · Collaboration · Access · Opportunity

Every week, colleagues make science learning stronger through shared planning, preparation, feedback, curriculum expertise, inclusive teaching, and opportunities that extend beyond the classroom. These ten recognitions capture only a small part of the generosity that makes New Bedford High School Science a true team.

01 Chemistry Collaboration

Ms. Jayni Gryss, Ms. Samantha Roche & Ms. Margaret Henning—thank you for lesson collaboration, pacing conversations, and sharing plans across PLC groups.

Shared expertise creates clearer, more coherent learning experiences.

02 Biology Planning & Benchmarks

Ms. Grace Stevenson, Ms. Seannine Tveit & Biology colleagues—thank you for lesson-sequence planning, standards-aligned adaptations, and writing benchmarks.

Common evidence helps teams understand student thinking across a course.

03 Chemistry Materials Readiness

Mr. Brian McCauley & Mr. Tyler Gilbert—thank you for reviewing, organizing, and managing Chemistry materials and supplies.

Prepared materials protect more time for student sense-making and laboratory work.

04 SPED Co-Teaching Partnership

Mr. Jacob Ramos, Ms. Cykka Lovett, Mr. Richard Dwinell, Mr. Mathew Golden & Ms. Leech—thank you for continuous collaboration and student support.

Shared routines and flexible access help every learner participate in rigorous science.

05 Feedback Culture

Thank you to the whole Science Department for welcoming constructive feedback for students, peers, and leadership.

Reflection and follow-through help us strengthen instruction together.

06 Newcomer Supports & Access

Ms. Lisa Richard, Ms. Seannine Tveit, Ms. Cassidy Barrett & colleagues—thank you for adapting lessons and sharing supports for newcomer students.

Clear language, visuals, and structured talk make science more accessible to all.

07 Genetics & Whale Watch

Ms. Kaelyn Dahl—thank you for refining the Genetics curriculum and planning the Whale Watch experience.

Curriculum stewardship and field learning connect science ideas to authentic questions.

08 Cuttyhunk Field Studies

Ms. Patricia Gifford—thank you for the continuing planning, communication, and care behind Cuttyhunk field trips.

Field study gives students a living context for evidence, systems, and inquiry.

09 Learning from One Another

Thank you to colleagues who share strategies, questions, materials, and encouragement across our department.

Professional generosity turns a challenge into a stronger next step.

10 Science Department Family

Thank you to every member of our department for welcoming students, maintaining high expectations, and creating meaningful experiences.

When we work as a team, every learner has a stronger opportunity to thrive.

WHAT OUR DEPARTMENT MAKES POSSIBLE

When teachers prepare together, share feedback, strengthen access, and open doors to authentic science experiences, students encounter a department that is ready to support both curiosity and high expectations.

October 5–9, 2026

Shared Practices, Stronger Learning

Department-wide routines that support student thinking in every science classroom

OUR COMMON PURPOSE

Make Science Learning Visible, Accessible & Actionable

Whether students are investigating a phenomenon, planning an experiment, building a model, analyzing data, writing an explanation, or completing a project, our shared work is to make the next step in learning clear and achievable.

> ***"A strong science department shares routines that invite every student to observe, ask questions, use evidence, explain their thinking, and improve."***

01 Start with a Clear Learning Target

Students benefit when they understand the scientific idea or practice they are developing and what strong work will show.

In practice: Name the target before the task, then invite students to use it as a check during discussion, investigation, or revision.

- State the knowledge or skill in student-friendly language.
- Connect the task to a meaningful science question.
- Return to the target during reflection or revision.

02 Plan for Evidence, Not Just Completion

In every course, teachers can identify where student thinking will become visible: a model, data table, discussion, explanation, calculation, or product.

In practice: Decide in advance which student product will show the scientific reasoning—not only whether an assignment was completed.

- Decide what evidence will be collected.
- Notice patterns, not only correct answers.
- Use what students produce to shape the next lesson.

03 Feedback That Prompts Student Action

Feedback is most powerful when it keeps the cognitive work with the learner and provides a manageable next move.

In practice: Use a prompt such as "Add one data point that strengthens your claim, then explain why it matters."

- Connect the comment to a goal or success criterion.
- Name one next step: revise, add evidence, clarify, compare, or explain.
- Build in time for students to respond and improve.

04 Access Through Multiple Pathways

Rigorous science becomes more accessible when students can encounter an idea through language, visuals, models, investigation, discussion, and writing.

In practice: Pair a visual representation with structured partner talk before asking students to write an evidence-based explanation.

- Pre-plan key vocabulary, representations, and supports.
- Structure partner or group talk around evidence.
- Maintain high expectations while varying the pathway.

THE DEPARTMENT LEARNING CYCLE

Plan together
Clarify the learning experience, materials, and supports.

Make thinking visible
Use tasks that reveal what students understand and can do.

Respond with feedback
Offer a specific next step that students can use.

Adjust together
Use evidence and colleague insight to strengthen the next learning experience.

Our shared commitment is simple: create rigorous, welcoming, and evidence-rich science learning so every student can participate, revise, and grow.